

Seeking validity with psychedelics

Psychedelics fracture fundamentals, prism like, bringing the ordinary processes of mental health science into view differently. They offer hope, a mysterious appeal of their ability to create perceptual and emotional shifts. But psychedelics research challenges scientific validity, emphasising how clinical trials require a spectrum of conditions to be met to generate knowledge. As van Elk and Fried describe, one of these conditions is the internal validity of a trial—whether the treatment effect was due to the treatment itself. In psychedelic trials, internal validity is compromised by various factors—the absence of control groups, the selection bias involved in people who enrol, and challenges with treatment blinding—that prevent researchers controlling for the effects of expectation and placebo. These factors and trials' small size also limit external validity, making it difficult to extrapolate trial results to real-world settings. Another difficulty is construct validity, since it is unclear how these treatments work, in terms of the drug's mode of action in the brain, and because treatments often involve a poorly described and understood psychotherapy component.

The *Lancet Psychiatry* published one of the earliest of the recent psychedelic trials, on psilocybin, but very little since then, because most of the studies we receive lack validity in some way. Encouragingly, trials are underway that address the breaking blind problem—eg, by using low-dose psychedelics as an active control group and assessing blinding efficiency. Placebo and nocebo effects can be reduced by assessing participants' and researchers' expectations, and conveying more realistic expectations. Studies could recruit participants with no previous psychedelic experience to address selection bias. As with all clinical trials, those of psychedelics should be adequately powered and assess long-term effects. Adverse effects should be systematically monitored and reported, and psychological components should be systematically evaluated and described.

The psychotherapy component comes with its own challenges. Psychedelic science sits at the meeting point of two often separate approaches to mental health—studying effects of drugs and psychotherapy. Psychedelic therapy has typically been conceptualised as psychedelic-assisted psychotherapy, the drug embedded in a context with elements of psychotherapy, such as establishing a strong therapeutic alliance, fostering self-regulation, and

building participants' capacity to navigate distressing experiences. Gerhard Gründer and colleagues argue that the psychedelic drug and psychotherapy should not be viewed as separable, describing the importance of the psychotherapeutic context in enabling psychedelic states to facilitate corrective experiences.

Whatever the stance on context-drug interaction, the context needs to be clearly described, as highlighted by the European Medicines Agency in a Comment on what is needed from psychedelics research. In a new systematic review, Carolina Seybert and colleagues provide useful analysis of how psychological interventions in psychedelic treatments are reported. They found great variability in the number and duration of sessions reported, the qualification and training of therapists, and assessment of therapist adherence to therapy. Although some studies reported psychotherapeutic modalities, the characteristics of interventions were often not described in detail, which shows what future trials need to improve.

Psychedelics also offer a chance to reconsider how drug and psychotherapy trials can fit together. Ioana Cristea and colleagues call for the use of evidence-based psychotherapies in psychedelic trials to distinguish the effect of the psychotherapy component from that of the psychedelic and to facilitate comparisons among trials. More broadly, they propose that psychiatric drug trials should not only use placebo comparisons, but also use evidence-based psychotherapy as an active comparison group. They note that trials of psychiatric drugs often define conditions such as treatment-resistant depression in terms of unsuccessful drug treatments only, even though psychotherapies are established treatment options for many mental disorders. This separation of the analysis of drug and psychological treatments is tightly bound with regulatory decisions, since public agencies regulate drugs, but psychotherapies are largely unregulated. Bringing together the investigation of psychiatric drugs and psychotherapy could transform the regulation of treatments for mental disorders. Psychedelics provoke reframing, and the integration of drugs and psychotherapy could be a useful step in mental health science. Overall, psychedelics trials should be better designed to meet validity requirements so that the potential of psychedelics can be more realistically and safely determined. ■ *The Lancet Psychiatry*



For more on **validity in psychedelic science** see *Ther Adv Psychopharmacol* 2023; **13**: 20451253231198466

For the **psilocybin trial** see **Articles** *Lancet Psychiatry* 2016; **3**: 619–27

On the **inseparability of psychedelics from psychotherapy** see **Personal View** *Lancet Psychiatry* 2024; **11**: 231–36

For the **European Medicines Agency** see **Comment** page 7

For the **review of psychological interventions in psychedelic treatments** see **Review** page 54

For more on **psychotherapy in drug trials for mental disorders** see **Comment** *Lancet Psychiatry* 2022; **9**: 934–36